VIPP Report

DeepInfer Prostate MRI Segmentation Evaluation

PROMISE12 Dataset

Prepared by

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Report

Introduction

In this report, I evaluated the `prostate-segmenter` model from the Docker image `deepinfer/prostate` for prostate MRI segmentation on the PROMISE12 dataset. The original data were provided as MetaImage `.mhd/.raw` files and were converted to `.nrrd` format for inference. Before evaluation, the converted `.nrrd` files were verified against the original `.mhd` files so that image size, voxel spacing, origin, and orientation metadata matched correctly.

A description of the model can be found at: https://www.deepinfer.org/models/prostate-segmenter/

The dataset description can be found at: https://promise12.grand-challenge.org/Details/

The evaluation was carried out as a whole-gland segmentation task using the binary prostate masks as ground truth. DeepInfer inference was run in the PROMISE12 domain using the accurate ensemble mode. Although Docker GPU passthrough was available on the workstation, the model container uses an older TensorFlow runtime that did not expose a usable GPU device inside the container, so the full evaluation was completed on CPU.

At the end of the evaluation, I obtained an average whole-gland Dice score of **0.8486** across all 50 PROMISE12 cases. Geometry validation passed for all evaluated cases.

Qualitative Comparison

The following screenshots are intended to visually compare the provided ground-truth annotation and the model prediction. In these screenshots, the annotation should be shown in **green**, while the model output should be shown in **red**.

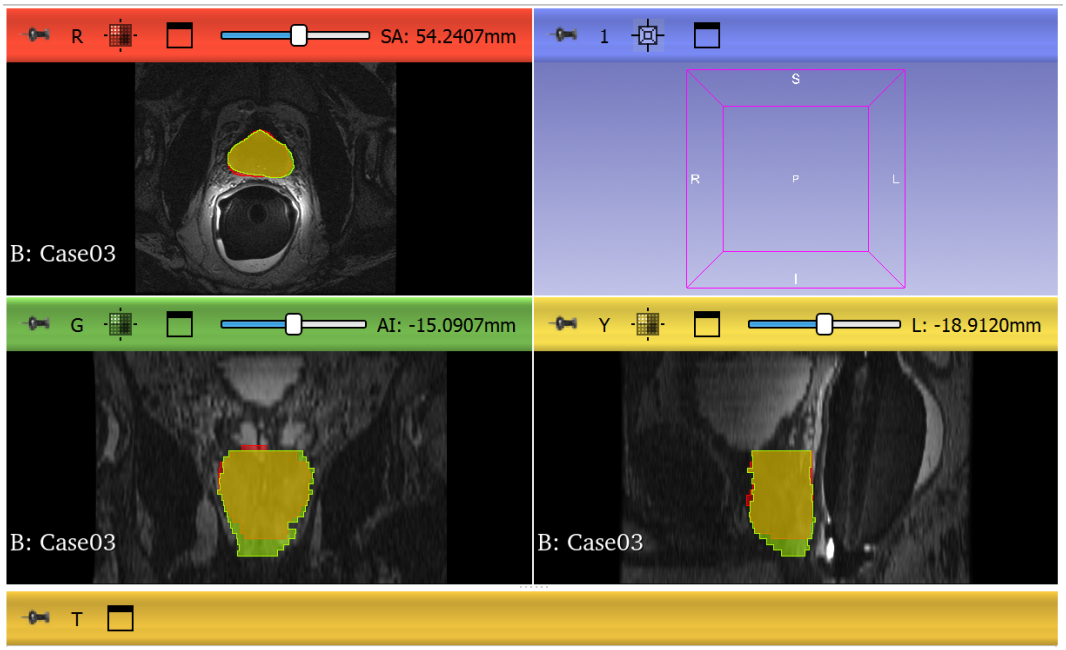

Figure 1: Comparison between ground truth and model output. Model output in red, ground truth in green.

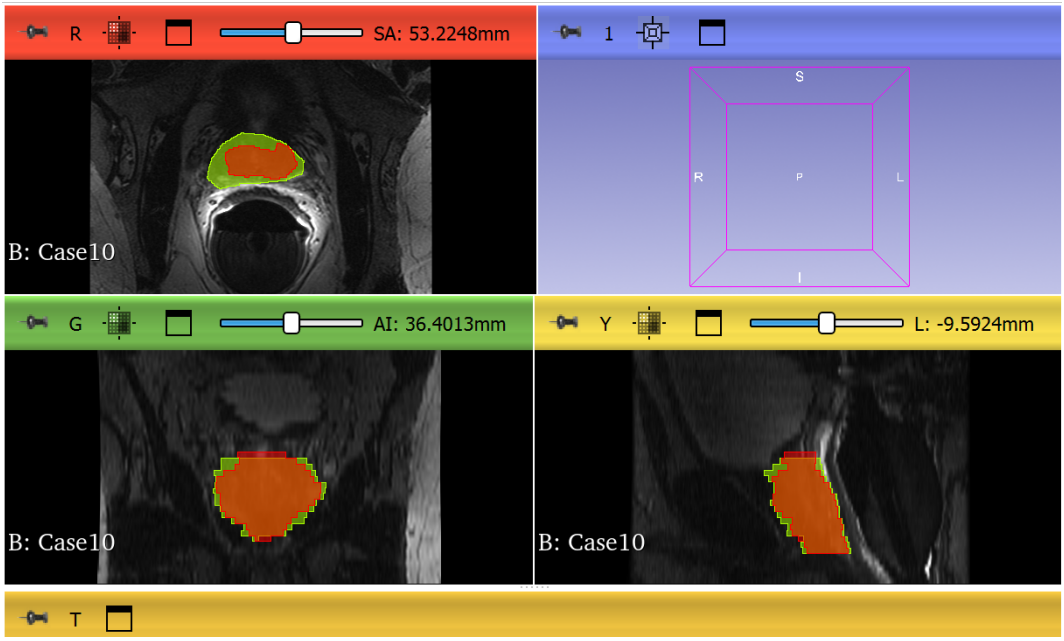

Figure 2: Comparison between ground truth and model output. Model output in red, ground truth in green.

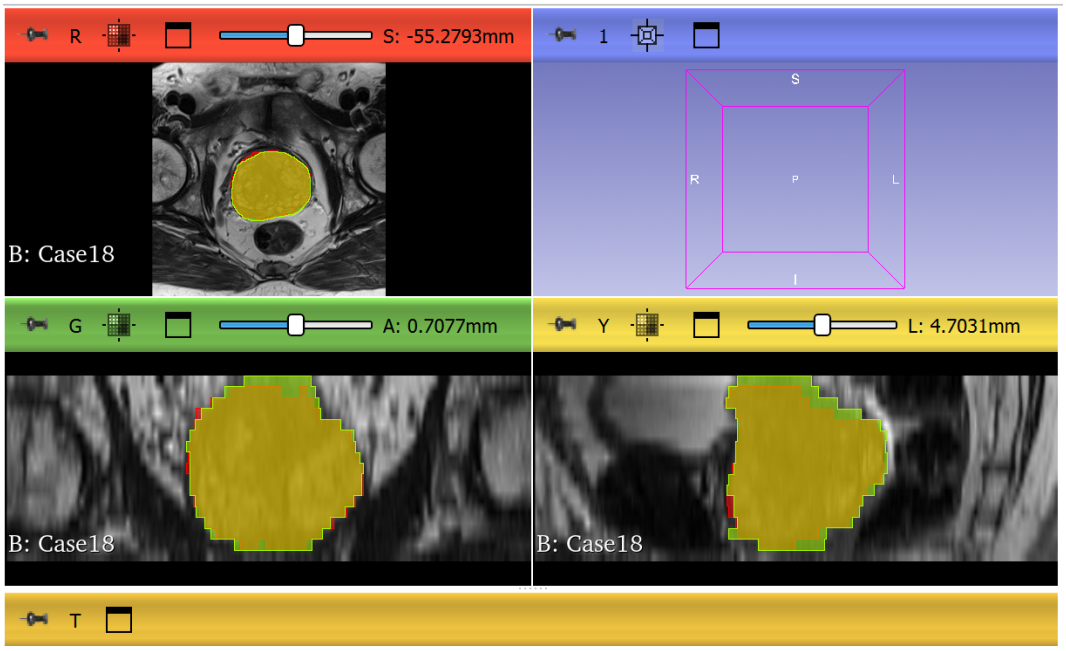

Figure 3: Comparison between ground truth and model output. Model output in red, ground truth in green.

Quantitative Results

Table 1: Dice score results obtained on the PROMISE12 dataset. Since the ground truth consists of binary whole-gland annotations, zonal Dice scores are not applicable.

Case	Mode	Ground Truth	Whole Gland Dice
Case00	whole-gland	binary GT	0.8205
Case01	whole-gland	binary GT	0.9587
Case02	whole-gland	binary GT	0.3926
Case03	whole-gland	binary GT	0.8985
Case04	whole-gland	binary GT	0.8379
Case05	whole-gland	binary GT	0.6841
Case06	whole-gland	binary GT	0.3960
Case07	whole-gland	binary GT	0.9104
Case08	whole-gland	binary GT	0.9281
Case09	whole-gland	binary GT	0.3568
Case10	whole-gland	binary GT	0.8731
Case11	whole-gland	binary GT	0.9156
Case12	whole-gland	binary GT	0.7271
Case13	whole-gland	binary GT	0.9311
Case14	whole-gland	binary GT	0.9587
Case15	whole-gland	binary GT	0.9490
Case16	whole-gland	binary GT	0.9470
Case17	whole-gland	binary GT	0.9289
Case18	whole-gland	binary GT	0.9412
Case19	whole-gland	binary GT	0.9413
Case20	whole-gland	binary GT	0.9364
Case21	whole-gland	binary GT	0.9351
Case22	whole-gland	binary GT	0.9584
Case23	whole-gland	binary GT	0.9677
Case24	whole-gland	binary GT	0.9636
Case25	whole-gland	binary GT	0.9493
Case26	whole-gland	binary GT	0.8872
Case27	whole-gland	binary GT	0.8489
Case28	whole-gland	binary GT	0.9464
Case29	whole-gland	binary GT	0.9630
Case30	whole-gland	binary GT	0.7676
Case31	whole-gland	binary GT	0.9473
Case32	whole-gland	binary GT	0.9542
Case33	whole-gland	binary GT	0.9172
Case34	whole-gland	binary GT	0.9540
Case35	whole-gland	binary GT	0.9471
Case36	whole-gland	binary GT	0.9416
Case37	whole-gland	binary GT	0.9247
Case38	whole-gland	binary GT	0.7366
Case39	whole-gland	binary GT	0.5036
Case40	whole-gland	binary GT	0.5335
Case41	whole-gland	binary GT	0.8576

Case	Mode	Ground Truth	Whole Gland Dice
`Case42`	whole-gland	binary GT	0.7658
`Case43`	whole-gland	binary GT	0.8230
`Case44`	whole-gland	binary GT	0.7801
`Case45`	whole-gland	binary GT	0.8621
`Case46`	whole-gland	binary GT	0.8243
`Case47`	whole-gland	binary GT	0.8945
`Case48`	whole-gland	binary GT	0.9243
`Case49`	whole-gland	binary GT	0.9178
Dataset mean	whole-gland	binary GT	0.8486

Applicability to Other MRI Sequences

Based on the official DeepInfer model documentation, this model is intended for **pelvic axial T2-weighted MRI scans**. The DeepInfer `Prostate-Segmenter` model page describes the task as whole-gland prostate segmentation in pelvic axial T2-W MRI scans and lists the required input as `Pelvic T2-W MRI`. In the command-line guide, the `InputVolume` is also explicitly described as a *T2-Weighted Pelvic MRI*. In addition, the official PROMISE12 dataset description states that the training cases consist of *transversal T2-weighted MR images of the prostate*. Since the PROMISE12 domain of this model was trained on that dataset, the evidence supports use on T2-weighted images rather than on other MRI sequences.

Therefore, the answer is effectively **no** for supported use on non-T2 sequences such as `ADC`, `DWI`, or `T1`-weighted MRI. The container may still accept another image file as input, but this would be an **out-of-domain use case** and there is no official evidence in the model documentation that performance on other sequences is valid or reliable. In practice, the model should be treated as a **T2-only segmentation model** unless it is retrained or separately validated on the alternative sequence.

Relevant sources: DeepInfer Prostate-Segmenter model page, PROMISE12 challenge data description.

Discussion

Overall, the model achieved strong whole-gland overlap on most PROMISE12 cases, with several cases reaching Dice scores above 0.94. The best-performing case was `Case23` with a Dice score of 0.9677. The weakest result was `Case09` with a Dice score of 0.3568, and other difficult cases included `Case02`, `Case06`, `Case39`, and `Case40`. These lower scores indicate that the model can substantially under-segment or mis-segment the gland on certain scans.

Despite these failure cases, the overall mean Dice of 0.8486 suggests that the DeepInfer model provides reasonably strong whole-gland prostate segmentation performance on the PROMISE12

dataset. Since all geometry checks passed between the original `.mhd` files, the converted `.nrrd` files, and the predicted segmentations, the reported scores reflect model performance rather than file-conversion mismatch.